

Online Experiments for Language Scientists

Lecture 2: Crowdsourcing

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What you will have read for today...

Reading tasks for this week

There are several things to read/look at this week, to give you a feel for some of the issues around online vs lab data collection, demographics of online populations, what online experiments look like, and what a crowdsourcing site looks like from the participant's perspective.

Read:

- » [The wikipedia page explaining what MTurk is.](#)
- » [Stewart, N., Chandler, J., & Paolacci, G. \(2017\). Crowdsourcing Samples in Cognitive Science. *Trends in Cognitive Sciences*, 21, 736-748.](#)

Also read *at least one of*:

- » [Monroe, R. et al. \(2010\). Crowdsourcing and language studies: the new generation of linguistic data. In *Proceedings of the NAACL HLT 2010 Workshop on Creating Speech and Language Data with Amazon's Mechanical Turk*, pages 122–130.](#)
- » [Pavlick, E. et al. \(2014\). The Language Demographics of Amazon Mechanical Turk. *Transactions of the Association for Computational Linguistics*, 2, 79-92.](#)
- » [A blogpost listing some downsides of MTurk](#), although note that this is written by people at Prolific, who are an MTurk competitor! Doesn't mean some of the points aren't valid though.
- » [A 2018 article in The Atlantic on worker exploitation on MTurk](#), although remember that we at least control what we pay workers and how we treat them.

It might also help to have a look at [a video of what the MTurk site looks like from a worker's perspective](#) - NB this video dates from December 2020. I'll show you around Prolific in class.



Online experiments & Crowdsourcing

- When an experiment runs in a browser, participants can use their own device, when/wherever works for them.
 - Send a link to your pals
 - To all the students in a class
 - On a Reddit forum
- Or advertise on crowdsourcing sites
 - Websites with populations of “workers”
 - Participants who do online tasks for money

MTurk and Prolific

Amazon Mechanical Turk

<https://www.mturk.com>

- Designed for crowdsourcing **anything**
- Very light touch
- More US-based participants?
- Interface is pretty horrible (particularly for experimenter) but has a powerful API
- More chaotic, worse data/exploitation

Prolific (formerly “Prolific Academic”)

<https://www.prolific.com>

- Designed for scientific data collection
- Heavier vetting of participants
- More UK/EU participants?
- Nicer web interface, recently added API
- Maybe better-behaved participants

A look around Prolific

- From a participant perspective
- From an experimenter perspective



Waiting for studies

Please check your [About you](#) page periodically to ensure everything is up to date.

Note: you do not need to refresh the page. Studies will appear here automatically. [Disable animation](#)



Crowdsourcing pros and cons...

Pros

- **Large samples, fast**
- Access different populations
- + for replicability

Cons

- Expensive (**not** cheap)
- Lack of control
- Encourages dumb experiments?
- - for replicability

Pro: large samples, *fast*

MTurk and Prolific both have large active populations of workers/participants (100,000s of registered people)

- Not everyone is active all the time
 - Estimating Mturk population size is complicated (see e.g. Difallah et al., 2018)
 - Prolific gives you an estimate of available and active population size

In practice, you can recruit **100s/1000s of participants in days.**

Difallah, D., Filatova, E., & Ipeirotis, P. (2018). Demographics and Dynamics of Mechanical Turk Workers. In *Proceedings of WSDM 2018: The Eleventh ACM International Conference on Web Search and Data Mining*.

<https://doi.org/10.1145/3159652.3159661>

Pro: access different populations

Typical lab-based studies sample from:

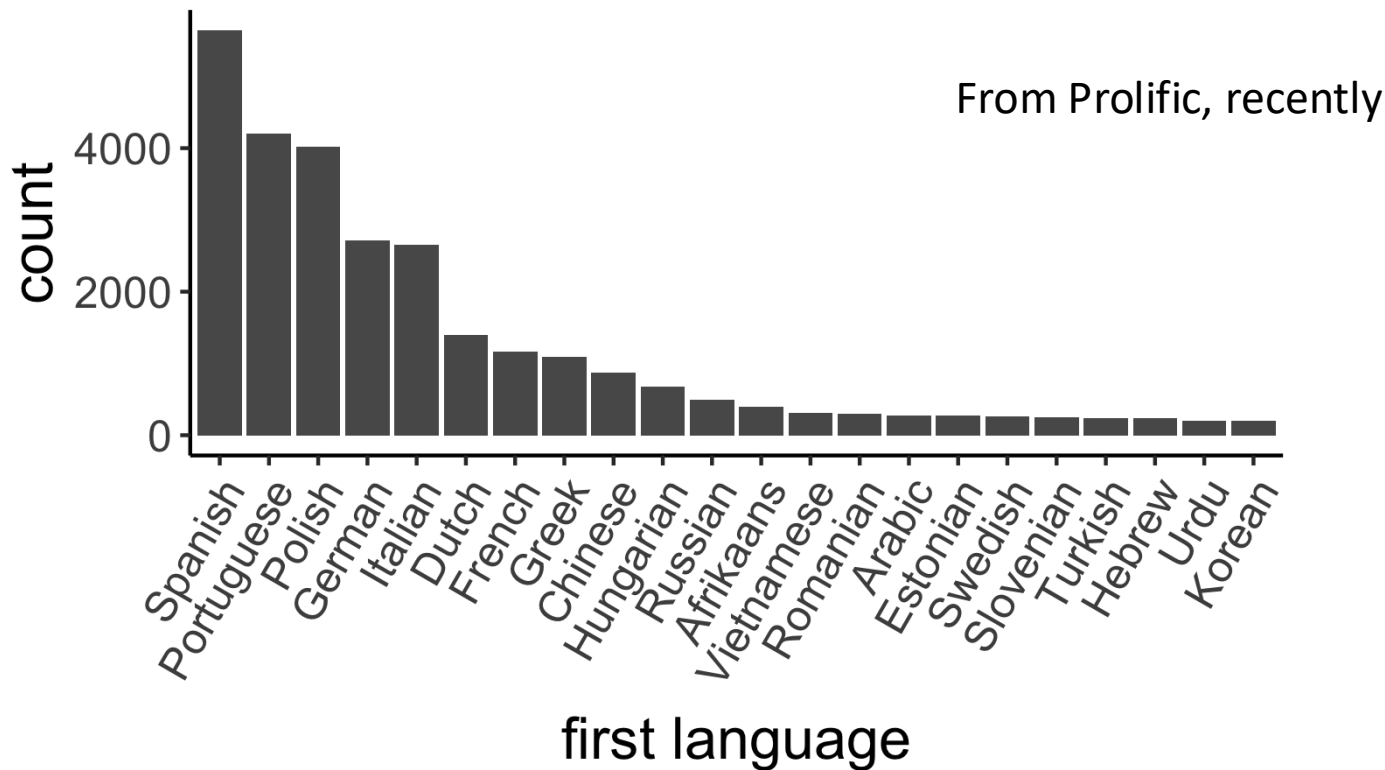
- University student population
- Mostly undergraduates
- Mostly young
- All highly educated
- Here, mainly native English speakers (obviously varies between unis)

If you want to access a different population, crowdsourcing might let you do that

workers	quality	speed	
many	high	fast	Dutch, French, German, Gujarati, Italian, Kannada, Malayalam, Portuguese, Romanian, Serbian, Spanish, Tagalog, Telugu
		slow	Arabic, Hebrew, Irish, Punjabi, Swedish, Turkish
	low or medium	fast	Hindi, Marathi, Tamil, Urdu
		slow	Bengali, Bishnupriya Manipuri, Cebuano, Chinese, Nepali, Newar, Polish, Russian, Sindhi, Tibetan

From Pavlick et al. (2014)

Pro: access different populations

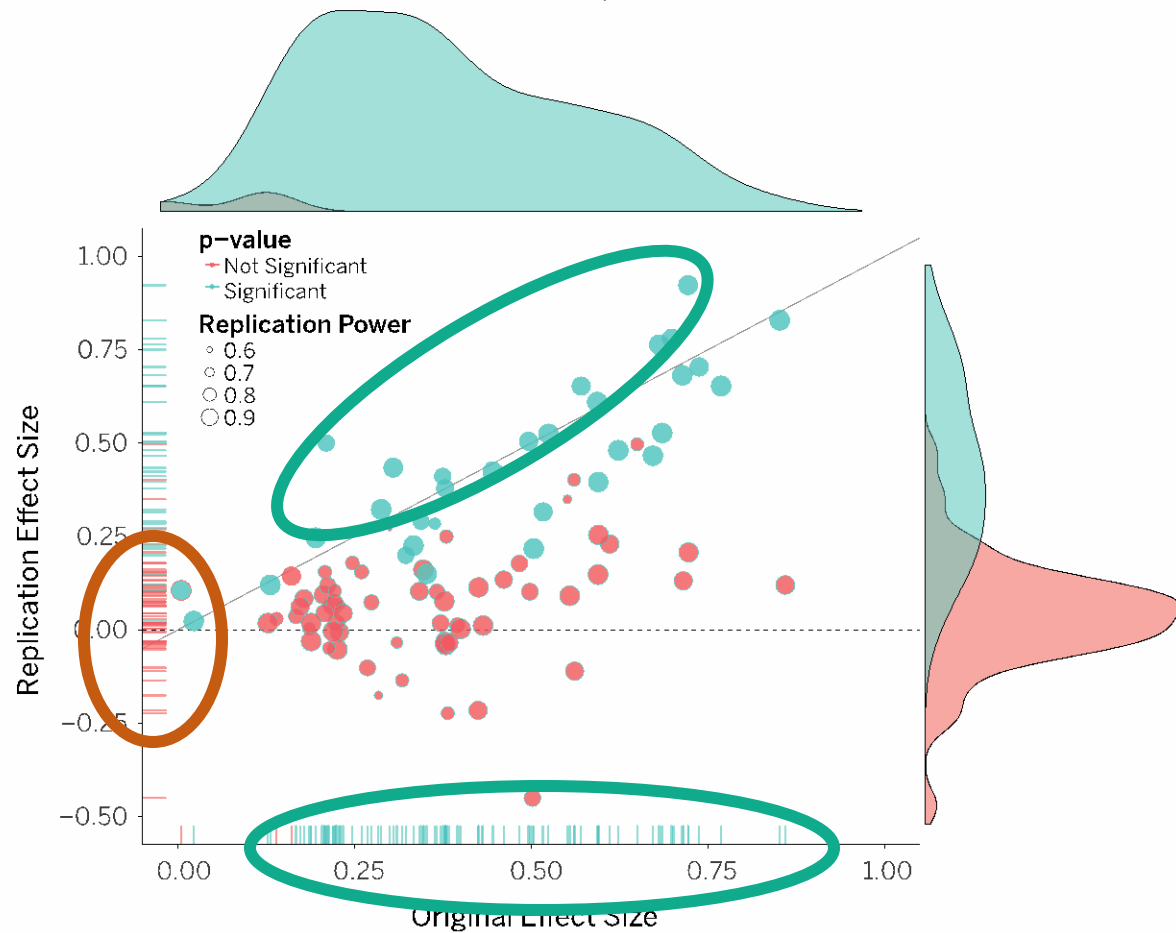


Pro: + for replicability

If you see a result in a scientific paper, can you assume that the effect they report is real and not just, e.g., a statistical fluke?

One way to check: *replication*

- Take someone else's experiment, replicate it, check you get the same result
- *Replication crisis*: many classic psychology experiments had small sample sizes, did NOT replicate!
- Crowdsourcing offers in principle much larger sample size



From Open Science Collaboration, 2015, *Science*

Pro: + for replicability

If you see a result in a scientific paper, can you assume that the effect they report is real and not just, e.g., a statistical fluke?

One way to check: *replication*

Multiple potential advantages for online data collection

- Large samples are easy, avoid small-sample experiments
- Collecting data is fast and easy; easier to replicate experiments
- Populations are shared (avoids e.g. “ah it’s because your population is different” responses to non-replication)

Con: **expensive**, *not* cheap

Mturk does not set minimum pay rates

Prolific does, but they are low (£6/hour)

But we should not be paying at those rates

It's unethical

It's exploitative

Current rates

The rates change on 1 April every year.

From April 2025, the rates are:

- £12.21 (aged 21 and over)
- £10 (aged 18 to 20)
- £7.55 (aged under 18)
- £7.55 (apprentice rate)

Con: **expensive**, *not* cheap

Mturk does not set minimum pay rates

Prolific does, but they are low (£6/hour)

But we should not be paying at those rates

It's unethical

It's exploitative

Additionally...

- Mturk and Prolific charge fees: 20-40% on top of participant payment
- Plus, bigger sample sizes



£2 for a 10
minute
experiment

Your behaviour on crowdsourcing platforms

- Pay fairly
- Estimate task durations accurately, and track it!
- Reject infrequently, if at all
- Respond promptly and politely to questions, don't get into arguments
- Test and test before putting your study up
- **If in doubt, pay out**

A

AfterDarkMark

Average Pace

A/B Testing - \$0.40



Underpaid



Unrated



Approved

\$4.36 / hour

00:05:30 / completion time

Pros

ymmv, but still probably underpaid, even if you fly through.

master's qual

Cons

Newer requester, though showing 99% approval on limited info, as of this post.

Not sure if it can be done that much faster, as a couple questions seemed kind of obvious to have 'correct' answers, so I assume those were quality/ attention type checks.

Pays poorly.



r/ProlificAc



Search in r/ProlificAc



u/cy_kelly • 13 hr. ago



I don't think I've ever been less qualified for a study

Discussion



A Study on Experiencing "Too Much Passion" for Your Work?

By hbs.edu

\$1.80 • \$12.00/hr

9 mins

233 places

Survey

In-study screening



86



10



Share

Microsoft Teams

Con: lack of *control*

In a normal lab study, you:

- Interact with your participants
 - ✓ Check they are indeed e.g., a human who speaks English natively
- Control where they are
 - ✓ Quiet space, same equipment for everyone
- Monitor them as they participate, and they know this
 - ✓ No falling asleep!

With crowdsourced participants participating remotely, **none of these things are true!**

- Consequently, experiments need to be designed to handle this

Compensating for lack of control

- Add checks on who the participants are, e.g.:
 - native language checks
 - instruction comprehension checks
- Add attention checks during the task, identify (and reject?) people who are not attending or who are responding randomly
- How? Ideas?



wug

derf



nej

derf



blik

keys

Compensating for lack of control

- Add checks on who the participants are, e.g.:
 - native language checks
 - instruction comprehension checks
- Add attention checks during the task, identify (and reject?) people who are not attending or who are responding randomly
- Can you make it easier to pay attention than not?



wug

derf



nej

derf

....wait 3 seconds...

Compensating for lack of control

- Add checks on who the participants are, e.g.:
 - native language checks
 - instruction comprehension checks
- Add attention checks during the task
- Can you make it easier to pay attention than not?
- Make the experiment short and fun! Most tasks on these platforms are pretty dull.

Player: PSSH. Good luck hitting me. My AC is incredible.

DM:



Final note: comparability with lab data

No hard constraints, but because of the lack of control, stuff that works best:

- Constrained and low-effort responses, e.g.:
 - One-off decisions (i.e. not involving complex integration of info)
 - Few restricted choices per trial (not e.g. open-ended typing)
 - Short experiments (a few minutes rather than an hour)

Can you investigate the questions you want using these sorts of methods?

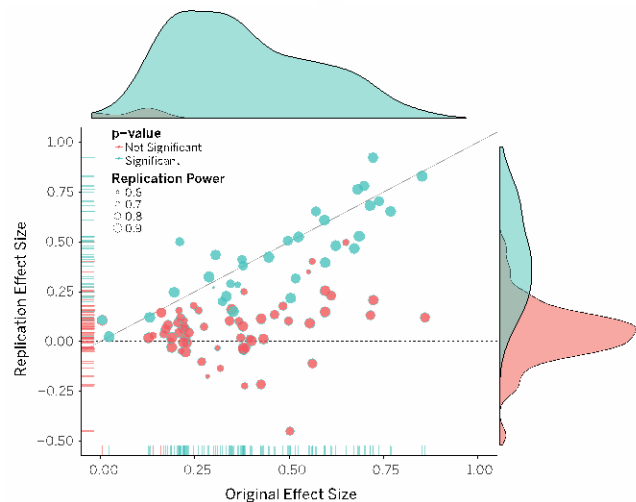
Con: – for replicability (?)

If you see a result in a scientific paper, can you assume that the effect they report is real and not just, e.g., a statistical fluke?

One way to check: *replication*

Potential risk of online data collection:

- Collecting data online is fast and easy
- Could run lots of experiments, and just publish the ones that “work” (cf. “the file drawer problem”)



Con: – for replicability (?)

People often want to know if crowdsourced data is like lab data (i.e. do effects shown in the lab replicate online?)

- Lab data as a “gold standard” due to higher levels of control
- Or just because the effect you are interested in has only been shown in the lab

We'll see numerous papers making direct comparisons, or replicating lab results with crowdsourced populations (e.g. from this week's set readings, Monroe et al., 2010)

Time for Q&A/discussion on this week's readings?

Next up

Thursday, 12:10: lab

- More basics of jsPsych and javascript
- Come prepared!

Next Monday: grammaticality judgments

- Do the reading beforehand!